Mission Veteran AI — Architecture

ibm/granite-4-h-small · watsonx Orchestrate · watsonx Governance · Terraform-provisioned · 12 RAG sources

USER

Veteran Career Navigator — Chat UI

watsonx Orchestrate webchat · PII reminder panel · session-scoped, no data persistence

Inputs: service branch · specialty code (MOS/AFSC/rating) · service summary · accomplishment bullets · target career path · job description (paste from LinkedIn / Indeed / USAJobs)

GUARD

PII Guard (regex) + Content Safety Monitor **Governance**

Layer 1 — `_common.contains_pii()`: SSN · DoD ID · email · phone · address pattern matching — hard-blocks before LLM call | Layer 2 — watsonx Governance content safety monitor: ML-based PII, hallucination detection (≥80% grounding threshold), harmful content — realtime, every inference payload

ORCH

Coordinator Agent

watsonx Orchestrate · ibm/granite-4-h-small · style: react_core

`{service_branch, specialty_code, service_summary, bullets, career_path, job_description}`

Gated 5-step pipeline · delegates to 4 sub-agents · enforces PII reminder · blocks export until gate OPEN

📦 Terraform: watsonx Orchestrate service instance · ibm module v1.2.1

External APIs

O*NET Crosswalk API
O*NET Occupations API
O*NET Work Activities
BLS OOH API
USAJobs API ⏳
CareerOneStop ⏳

Referral Resources

Hire Heroes USA
LinkedIn Jobs
Indeed

ibm/granite-4-h-small (all sub-agents)

AGENTS

BulletTranslator

Military bullet → civilian ATS bullet

toolkit: `bullet_translator`

`translate_bullet()`

Prompt: "Do not invent facts not present in bullet or metrics"

Governance logged

SkillMapper

MOS/AFSC/rating → civilian skill inventory

toolkit: `skill_mapper`

`map_military_skills()`

NEW Evidence field per skill (verbatim quote)
No forced skill count — direct evidence only
O*NET grounding · `evidence_sourced_count`

Bias monitor: service_branch

GapAnalyzer

Skill inventory → gap report + recommendations

toolkit: `gap_analyzer`

`analyze_skills_gap()`

coverage_pct · SkillsBuild courses · SkillBridge programs · Hire Heroes USA referral

Bias monitor: career_path

PlanBuilder

JD analysis → tagged resume draft → export gate

toolkit: `plan_builder`

`analyze_job_description()`

`build_resume_draft()`

`evaluate_export_gate()`

NEW Pass 1: draft + EVIDENCED/INFERRED/MISSING tags
NEW Pass 2: grounding check — downgrades unsupported EVIDENCED → INFERRED
NEW Gate blocks on `grounding_downgrade_count` + inferred + missing

Explainability monitor

GOV

WATSONX GOVERNANCE — AI FACTSHEETS & MONITORS

Bias & Fairness

service_branch fairness
career_path fairness
alert threshold: ±10 pp
daily evaluation

Model Drift

data drift: JS divergence
quality drift: coverage_pct
baseline: 30-day window
weekly evaluation

Content Safety

ML PII scan (input+output)
hallucination: ≥80% grounding
harmful content detection
realtime · every inference

Explainability

LIME per inference
skill attribution
coverage score drivers
on-request retrieval

4 AI use cases registered · factsheets.yaml + monitors.yaml · registered via governance/register_factsheets.py (post terraform apply)

📦 Terraform: `ibm_resource_instance "governance"` (service: aiopenscale) · GUID output → `GOVERNANCE_INSTANCE_GUID` env var

RAG

KNOWLEDGE BASES & RAG SOURCES (12 REGISTERED)

Static JSON KBs

onet_military_crosswalk.json
onet_occupation_skills.json
skillsbuild_courses.json
skillbridge_programs.json

loaded via
`shared/_common.py`
annual / quarterly refresh

Prompt Templates

prompt_templates.json

O*NET Crosswalk

RAG-001 · Dept. of Labor
30 MOS/AFSC records
Annual · Static JSON

O*NET Occ. Skills

RAG-002 · Dept. of Labor
Skills · wages · outlook
Annual · Static JSON

O*NET Work Activities

RAG-008 · Dept. of Labor
Embedded in occ. skills
Annual · Partial embed

IBM SkillsBuild

RAG-003 · IBM
20 courses · credentials
Quarterly · Static JSON

DoD SkillBridge

RAG-004 · Dept. of Defense
12 programs
Quarterly · Static JSON

BLS OOH

RAG-005 · BLS
Wages · job outlook
Annual · Embedded

USAJobs API ⏳ pending

RAG-006 · OPM
Federal job postings
Real-time · API key req.

CareerOneStop ⏳ pending

RAG-007 · Dept. of Labor
Occ. profiles · training
Quarterly · API key req.

Job Postings (user)

RAG-009 · User-provided
LinkedIn / Indeed / USAJobs
Copy-paste per session

Hire Heroes USA

RAG-010 · 501(c)(3)
Referral · coaching
No API · DoD/VA partner

LinkedIn Jobs

RAG-011 · Referral
User-paste via RAG-009
No live API (ToS)

Indeed

RAG-012 · Referral
User-paste via RAG-009
API deprecated 2023

OUTPUT

Veteran Career Package — Gated Export

ATS civilian bullets · Skill inventory with evidence quotes · Skill gap report + coverage % · IBM SkillsBuild course recommendations · DoD SkillBridge program matches · Evidence-tagged resume draft (EVIDENCED / INFERRED / MISSING) · Grounding check warnings list · Hire Heroes USA referral

🔒 **Export gate**: BLOCKED until veteran resolves all grounding-check downgrades + INFERRED + MISSING items. Gate requires human sign-off before document leaves the system.

⬛ watsonx.ai + Orchestrate (LLM inference) 🟪 watsonx Governance 🟦 Static / embedded RAG source 🟧 Pending / live API RAG source ⬜ Referral-only RAG source

⬚ Terraform-provisioned resource 🟩 New in this release

Made with IBM Bob